ECE 329 | Fields and Waves I | Summer 19

Kudeki

Exam 2

Wed, July 10, 2019 — 11 AM - 11:50

Name:	Solution
Section:	

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

This is a open book exam but calculators are not allowed. You are allowed two formula sheets of 8.5 by 11 inch dimensions — both sides of the sheets may be used. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. **Coax capacitor problem:** A dielectric filled coax capacitor has a conductance per unit length of $\mathcal{G}$ =2 μS/m (μ=micro=10^{-6}). The conductivity of the filling is known to be $\sigma = 10^{-6}$ S/m.

   a) (7 pts) What is the capacitance per unit length $\mathcal{C}$ of the same capacitor if the filling permittivity $\epsilon = 2\epsilon_o$? Hint: free space permittivity $\epsilon_o \approx 8.854 \times 10^{-12}$ F/m.

$\mathcal{G} = \sigma(\mathcal{C}.F.) = 2\times10^{-6}\ \frac{S}{m}$

$\mathcal{C} = \epsilon(\mathcal{C}.F.) \Rightarrow \mathcal{C} = \frac{\epsilon}{\sigma}\mathcal{G} = \frac{2\times8.854\times10^{-12}}{10^{-6}}\ 2\times10^{-6}$

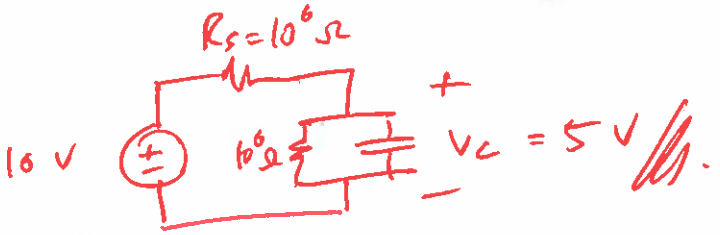
$= 4\times8.854\times10^{-12}$

$= 35.416\times10^{-12}\ \frac{F}{m}$

   b) (10 pts) A half a meter length of the coax capacitor described above is connected across a 10 V DC source with an internal resistance of $R_s = 10^6\,\Omega$ — what would be the capacitor voltage V_c in steady state?

$\frac{1}{R} = G = \mathcal{G}\ell = \left(2\times10^{-6}\ \frac{S}{m}\right)\left(\frac{1}{2}m\right) = 10^{-6}S \Rightarrow R = 10^6\Omega = 1M\Omega.$

$\uparrow R_s$ also



   c) (8 pts) If 10 V source in (b) were disconnected from the capacitor at time $t = 0$, at what later time t_1 would $V_c(t_1) = V_c(0)e^{-1}$? Hint: what is the decay time constant of the physical capacitor described above?

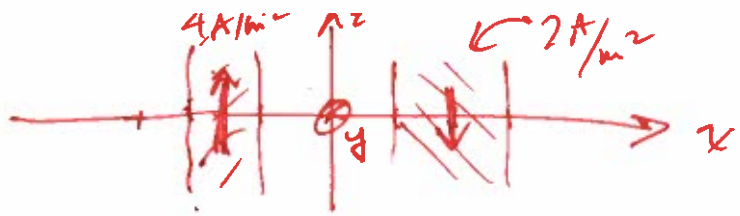
$\tau = RC$ ($R = 10^6\,\Omega$)

$C = \mathcal{C}\ell = 2\times8.854\times10^{-12}$

$= 17.708\times10^{-12}\,F$

$\leftarrow 17.708\times10^{-12}\,F$

$= 17.708\ \mu s.$



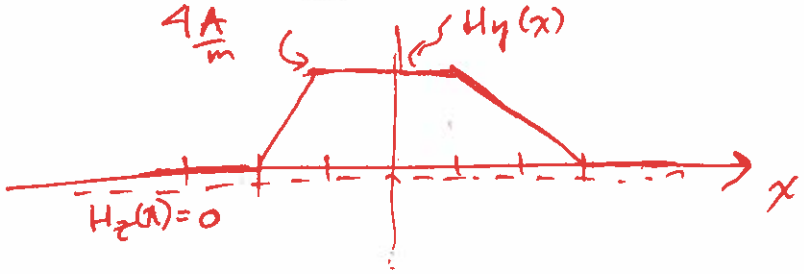
2. Two parts of this question are independent:

a) We have two infinite slabs of finite widths in the x-direction where a $\hat{z}$-directed constant volumetric current $\mathbf{J} = J_z\hat{z}$ flows. Slab 1 occupies the region $-2 < x < -1$ m and has $J_{1z} = 4$ A/m^2 while Slab 2 occupies the region $1 < x < 3$ m and has a $J_{2z} = -2$ A/m^2.

i. (5 pts) What is the magnetic field intensity vector $\mathbf{H}$ on the x-axis at $x = -3$, -1.5, 0, 1.5, 3 m?

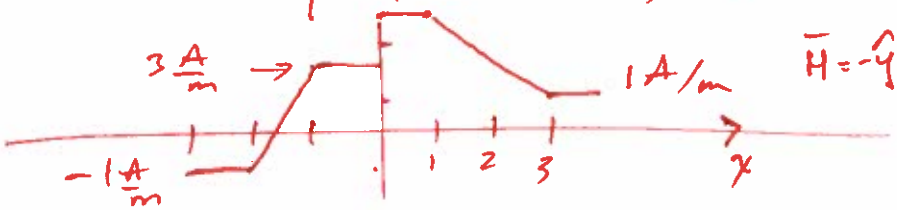
$\bar{H} = 0,\ \hat{y}\,2\frac{A}{m},\ \hat{y}\,4\frac{A}{m},\ \hat{y}\,3\frac{A}{m},\ 0$ @ $x = -3, -1.5, 0, 1.5, 3$ m

ii. (5 pts) Plot $H_x(x)$ and $H_y(x)$ as a function of x for $-3 < x < 3$ m — consistent with answers to part (i) — and justify your plot?



iii. (5 pts) How do the answers to parts (i) and (ii) are changed if a surface current sheet with $2\hat{z}$ A/m current density is placed on $x = 0$ surface?

→ subtract $\hat{y}\,1$ A/m from $x<0$, add $\hat{y}$ A/m for $x>0$



$\bar{H} = -\hat{y}\frac{A}{m}, \hat{y}\frac{A}{m}, \hat{y}\frac{4A}{m}, \hat{y}\frac{4A}{m}, \hat{y}\frac{1A}{m}$

@ $x = -3, -1.5, 0, 1.5, 3$

b) (10 pts) Consider magnetic vector potential $\mathbf{A}(x, y, z) = \hat{x}|z|$.

i. Show that $\nabla \cdot \mathbf{A} = 0$.

$\nabla \cdot A = \frac{\partial}{\partial x}A_x + \frac{\partial}{\partial y}A_y^{0} + \frac{\partial}{\partial z}A_z^{0} = \frac{\partial}{\partial x}|z| = 0$ ✓

ii. Determine $\mathbf{B}$ from $\mathbf{A}$.

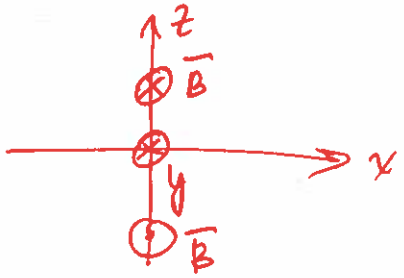
$$\bar{B} = \nabla \times \bar{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \pm z & 0 & 0 \end{vmatrix} = \hat{x}\,0 - \hat{y}\left(0 - (\pm 1)\right) + \hat{z}\,0 = \pm\hat{y}\ \frac{Wb}{m^2}$$

$z > 0$ ↑ , $z < 0$ ↓

iii. What would be $\mathbf{J}(x, y, z)$ that would produce $\mathbf{B}$ of (ii)?

$\bar{J} = \delta(z)\left(-\hat{x}\frac{2}{\mu_0}\right)\frac{A}{m^2}$

$\bar{J}_s = -\hat{x}\frac{2A}{\mu_0 m}$ at $z = 0$ surf.



$\bar{H} = \frac{\bar{B}}{\mu_0} = \pm\frac{1}{\mu_0}\hat{y}\frac{A}{m}$

3. **Solenoid and Faraday's law:** A wire loop C with a resistance of $R = 4\,\Omega$ is surrounding a tall solenoid of a circular shaped cross-section with 1 m^2 cross sectional area on the $z = 0$ plane. The solenoid carries a surface current density $J_\phi(t)$ in $\hat{\phi}$-direction coinciding with the direction of the fingers of a right hand when the thumb points in $\hat{z}$ direction. Answer the following questions:

a) (5 pts) If $J_\phi(t) = -2t$ A/m, what is the magnetic field intensity vector $\mathbf{H}$ within the solenoid at $t = 1$ s? Specify the magnitude, direction, and units of $\mathbf{H}$ properly.

$$\bar{H} = \hat{z}\, J_\phi(t) = \hat{z}(-2t)\ \frac{A}{m}$$

$$\bar{H}\Big|_{t=1s} = -2\hat{z}\ \frac{A}{m}$$

b) (5 pts) If $J_\phi(t) = -2t$ A/m, what is the magnetic flux $\Psi(t)$ linking the loop C surrounding the tall solenoid? Ignore the fringing fields of the solenoid in the flux calculation.

$$\bar{B} = \mu_0 \bar{H} = -2t\mu_0 \hat{z}, \qquad \Psi = (B_z)(1\,m^2) = -2t\mu_0 \text{ Wb.}$$

c) (5 pts) Given the flux of (b) what is the EMF of the wire loop C specified in the $\hat{\phi}$ direction?

$$\mathcal{E} = -\frac{d\Psi}{dt} = 2\mu_0 = 8\pi \times 10^{-7} \text{ V}$$

($\mu_0 = 4\pi \times 10^{-7}\ \frac{F}{m}$)

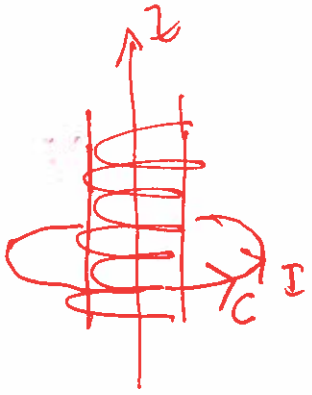
d) (5 pts) Given the EMF of (c) what is the loop current I around C specified in the $\hat{\phi}$ direction?

$$I = \frac{\mathcal{E}}{R} = 2\pi \times 10^{-7} \text{ A}$$

($R = 4\,\Omega$)

e) (5 pts) Discuss how current I direction from (d) complies with Lenz's law.

I direction is the same as Circulation direction.
With positive I, positive B_z is produced, that opposes the reducing B_z associated with $\Psi(t) = -2t\mu_0$ Wb.
Hence Lenz's law is consistent with our results.



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Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. **Capacitance problem:** Consider an air-filled parallel plate capacitor with capacitance C =8.854 pF (p=pico=10^{-12}).

   a) (5 pts) What would be C if the filling of the capacitor were altered by some dielectric with permittivity $\epsilon = 2\epsilon_o$? Hint: free space permittivity $\epsilon_o \approx 8.854 \times 10^{-12}$ F/m.

$C = \varepsilon_0 (G.F.) = \varepsilon_0 \rightarrow G.F. = 1\ m$

If $\varepsilon_0 \rightarrow 2\varepsilon_0 \Rightarrow C \rightarrow 2C$ $\therefore C = 2 \times 8.854 \times 10^{-12}\ F$

$= 17.708\ pF$

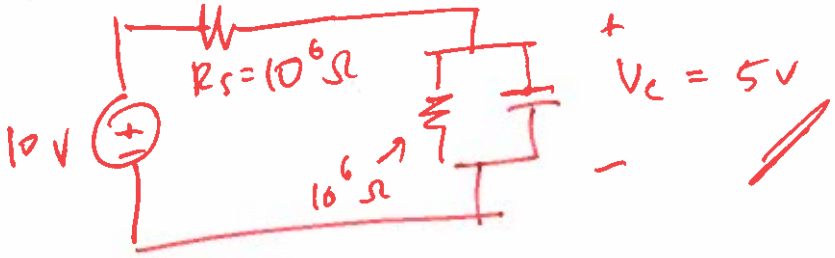
   b) (5 pts) What would be conductance of the same capacitor if the conductivity of the dielectric were $\sigma = 10^{-6}$ S/m?

$G = \sigma (G.F.) = 10^{-6}\ S$ (for 1 m)

$\sigma = 10^{-6}$ S/m

$\therefore R = \frac{1}{G} = 10^6\ \Omega$

   c) (7 pts) If the physical capacitor with the C and G given in (a) and (b) were connected across a 10 V DC source with an internal resistance of $R_s = 10^6\ \Omega$, what would be the capacitor voltage V_c in steady state.

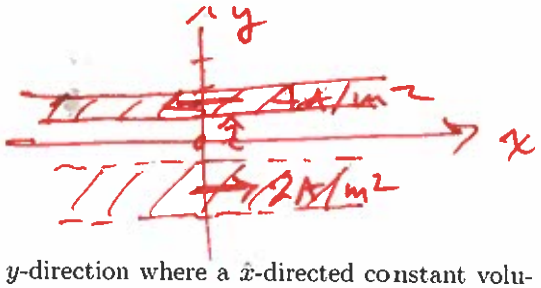


   d) (8 pts) If 10 V source in (c) were disconnected from the capacitor at time $t = 0$, at what later time t_1 would $V_c(t_1) = V_c(0)e^{-1}$? Hint: what is the decay time constant of the physical capacitor described above?

for $\varepsilon = 2\varepsilon_0$ case.

$\tau = RC = (10^6\ \Omega)(17.708 \times 10^{-12}\ F)$

$= 17.708\ \mu s$



2. Two parts of this question are independent:

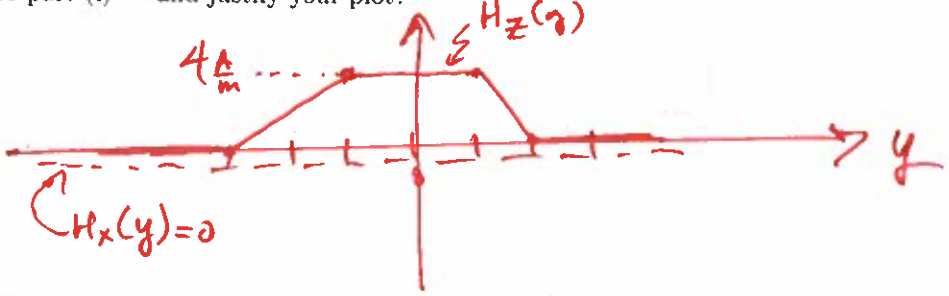
a) We have two infinite slabs of finite widths in the y-direction where a $\hat{x}$-directed constant volumetric current $\mathbf{J} = J_x\hat{x}$ flows. Slab 1 occupies the region $-3 < y < -1$ m and has $J_{1x} = 2$ A/m^2 while Slab 2 occupies the region $1 < y < 2$ m and has a $J_{2x} = -4$ A/m^2.

i. (5 pts) What is the magnetic field intensity vector $\mathbf{H}$ on the y-axis at $y = -3$, -1.5, 0, 1.5, 3 m?

$0,\ 3\hat{z}\frac{A}{m},\ 4\hat{z}\frac{A}{m},\ 2\hat{z}\frac{A}{m},\ 0$ @ $-3, -1.5, 0, 1.5, 3$ m

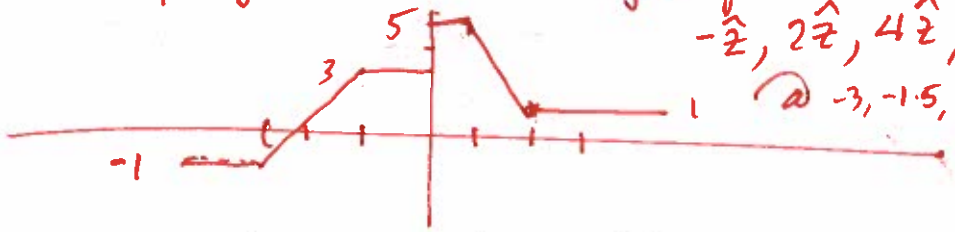
ii. (5 pts) Plot $H_x(x)$ and $H_z(x)$ as a function of y for $-3 < y < 3$ m — consistent with answers to part (i) — and justify your plot?

$H_x(y)$ $H_z(y)$



iii. (5 pts) How do the answers to parts (i) and (ii) are changed if a surface current sheet with -2$\hat{x}$ A/m current density is placed on $y = 0$ surface?

Add $-\hat{z}$1 A/m for $y>0$ & $+\hat{z}$1 A/m for $y<0$.



$-\hat{z}, 2\hat{z}, 4\hat{z}, 3\hat{z}, 1\hat{z}\ \frac{A}{m}$ @ $-3, -1.5, 0, 1.5, 3$ m

b) (10 pts) Consider magnetic vector potential $\mathbf{A}(x, y, z) = -\hat{y}|z|$.

i. Show that $\nabla \cdot \mathbf{A} = 0$.

$$\nabla\cdot\bar{A} = \frac{\partial A_x}{\partial x}^{0} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}^{0} = \frac{\partial}{\partial y}(-|z|) = 0.$$

ii. Determine $\mathbf{B}$ from $\mathbf{A}$.

$$\bar{B} = \nabla\times\bar{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & \mp z & 0 \end{vmatrix} = \hat{x}(0-(\mp 1)) - \hat{y}\,0 + \hat{z}\,0 = \pm\hat{x}\ \frac{Wb}{m^2}$$

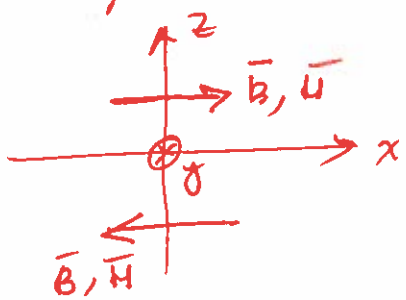
$z>0$: $+\hat{x}$; $z<0$: $-\hat{x}$

$$\bar{H} = \frac{\bar{B}}{\mu_0} = \pm\frac{1}{\mu_0}\ \frac{Wb}{m^2}$$

iii. What would be $\mathbf{J}(x, y, z)$ that would produce $\mathbf{B}$ of (ii)?

$$\bar{J} = \delta(z)\left(\hat{y}\,\frac{2}{\mu_0}\right)\frac{A}{m^2}$$

$$\bar{J}_s = \hat{y}\,\frac{2}{\mu_0}\,\frac{A}{m} \quad \text{at } z=0 \text{ surface.}$$



3. **Solenoid and Faraday's law:** A wire loop C with a resistance of $R = 4\,\Omega$ is surrounding a tall solenoid of a circular shaped cross-section with 1 m^2 cross sectional area on the $z = 0$ plane. The solenoid carries a surface current density $J_\phi(t) = Kt$ in $\hat{\phi}$-direction coinciding with the direction of the fingers of a right hand when the thumb points in $\hat{z}$ direction. Above K is a constant to be determined. Answer the following questions:

a) (5 pts) If current $I = 2\pi \times 10^{-7}$ A in loop C in the $\hat{\phi}$ direction, what is the EMF of the loop?

$\mathcal{E} = RI = (4\Omega)(2\pi \times 10^{-7}\,A) = 8\pi \times 10^{-7}\,V$ //.

b) (5 pts) Given the EMF of part (a), what is the magnetic flux $\Psi(t)$ linking the loop C.

$\mathcal{E} = -\frac{d\Psi}{dt} \Rightarrow \Psi = -8\pi \times 10^{-7} t$

c) (5 pts) Given the $\Psi(t)$ of part (b), what is $\mathbf{H}_z(t)$ within the solenoid? Ignore the fringing fields of the solenoid in this calculation.

$\Psi = B_z A \Rightarrow B_z = -8\pi \times 10^{-7} t$

$\bar{H} = \frac{\bar{B}}{\mu_0} = \frac{-8\pi \times 10^{-7} t\,\hat{z}}{4\pi \times 10^{-7}} = -2t\,\hat{z}\ A/s$

d) (5 pts) Given the $H_z(t)$ of part (b), what is the magnetic flux density $\mathbf{B}$ at the center of the solenoid at $t = 1$ s? Specify the magnitude, direction, and units of $\mathbf{B}$ properly.

$H_z(t=1s) = -2\ A/s$

$\bar{B} = \mu_0 \bar{H} \Rightarrow \bar{B}(t=1s) = (4\pi \times 10^{-7})(-2\hat{z}) = -8\pi \times 10^{-7}\hat{z}\ \frac{Wb}{m^2}$

e) (5 pts) Given the $H_z(t)$ of part (b), what is constant K used in $J_\phi(t) = Kt$ formula?

$H_z(t) = Kt = J_\phi(t)$

$\uparrow -2t \longleftrightarrow \therefore K = -2\,\frac{A}{m \cdot s}$

